Fiscal Year:	FY 2019	**Task Last Updated:**	FY 08/20/2019
PI Name:	Yarin, Alexander Ph.D.		
Project Title:	Stationary and Swing-Like Pool Boiling for Microgravity Applications for Cooling of High- Power Microelectronics		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ayarin@uic.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	(312) 996-3472
Organization Name:	University of Illinois at Chicago		
PI Address 1:	UIC Department of Mechanical & Industrial Engineering, (M/C 251), 2039 ERF		
PI Address 2:	842 W. Taylor Street		
PI Web Page:			
City:	Chicago	**State:**	IL
Zip Code:	60607	**Congressional District:**	7
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2009 ESA AO Biology on Sounding Rockets and Physical Sciences on All Platforms
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No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	McQuillen, John	**Contact Phone:**	216-433-2876
Contact Email:	john.b.mcquillen@nasa.gov		
Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX17AF33G		
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	NOTE: Continuation of "Spray Cooling in Flooding Regime: Pool Boiling on Nano-Textured Surfaces," grant NNX13AQ77G, Principal Investigator Alexander Yarin, PhD. The European Space Agency (ESA) recently initiated “Thermal Platform 1” research project, a ramification of DOLFIN, aiming research projects relevant for the work of the International Space Station (ISS) and for the further Space exploration. This document foresees that: "In the area of space applications and with the increasing size of space platforms or future exploration missions, large amounts of heat need to be dissipated by large space radiators. Efficient thermal control in space and the reduction of moving mechanical elements become of crucial importance and two-phase closed-loop systems can meet these requirements." The present project supported by NASA will address some of these issues and is planned to become an integral part of the ESA activities. The University of Illinois at Chicago (UIC) group		

Task Description:	supported by NASA in the framework of this project actively collaborated with the ESA supported group at the Technical University Darmstadt (Germany), where a group of 3 Master students, 1 PhD student, and 1 bachelor student were involved. Drop and spray cooling in such situations could offer an attractive alternative, and our previous work, supported by NASA as a part of participation in DOLFIN, was directed on the enhancement of drop cooling using nano-textured surfaces. In spray cooling with surface flooding, heat transfer occurs in the regime of pool boiling. The regimes of pool boiling can be divided on the boiling curve as free convection boiling, nucleate boiling, transition boiling, and film boiling. Among them nucleate boiling is considered to be the most desirable heat transfer regime, since it is associated with high heat fluxes at surfaces with low superheat temperature, which is attractive for such practical applications as cooling of electronics, boilers, nuclear and chemical reactors, and refrigeration systems. However, there exists a critical value of heat flux called critical heat flux (CHF) at which nucleate boiling transforms into film boiling. The latter involves heat transfer through coolant vapor rather than through liquid coolant itself. There were multiple efforts to increase CHF and boiling heat transfer coefficient (BHT), which is by definition the heat flux divided by the corresponding superheat temperature. In particular, boiling on roughened surfaces was found to increase BHT coefficient, which stems from an increased area density of nucleation sites. Microporous coatings to enhance pool boiling were also studied. In our preliminary work on pool boiling on nano-textured surfaces under full gravity conditions, in the ground experiments the nano-textured surfaces used were copper platelets covered with copper-plated electrospun nanofibers. In addition, for comparison, pool boiling on the corresponding bare copper surfaces was experimentally studied. The results revealed that the heat flux and heat transfer coefficient in boiling on the nano-textured surfaces were about 3-8 times higher than those on the bare copper surfaces. This stems from the fact that nano-textured surfaces promote bubble growth by increasing the average temperature of fluid surrounding growing bubbles, as our experimental and theoretical results show. Hence, nano-textured surfaces manufactured from copper-plated nanofibers facilitate bubble growth rate and, thus, increase bubble detachment frequency. The critical superheat temperature of nano-textured surfaces was about 10 degrees C lower than that on smooth surfaces. In addition, the heat flux on the nano-textured surfaces in transition boiling was significantly higher than on the bare copper ones, since the presence of nanofibers prevented bubble merging and delayed formation of vapor film. [ED. NOTE (12/2017): initial project from this work was " 'DOLFIN' DROP IMPACT ON POROUS NANO-TEXTURED SURFACES," (grant NNX10AR99G).]
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	[Editor's note (8/20/2019): compiled from PI's FY2019 annual progress report dated March 2019; received February 2019] We study the inception and propagation of vaporization fronts and transition to the critical heat flux (CHF) and film-boiling regime triggered by a steep, almost instantaneous increase in the heat release from a strip heater submerged in Novec 7300 liquid. The propagation path of the resulting vaporization front along the strip heater is measured and shown to be exponentially increasing in time, in distinction from the previous reports in literature claiming that the increase is only linear. Since the previous experiments employed such liquids as water, refrigerants, acetone, ethanol, and alkali metals, which possess relatively high latent heat of evaporation and thus require a relatively high power to be supplied, the heater burnout at the inception of the CHF and film boiling was too fast to allow for longer-time observations. Accordingly, the previous works observed only an extremely short-time asymptotics of the propagation process, which means the short-time expansion of the exponential function, which is linear. On the other hand, in the experiments with Novec 7300 liquid, the heater burnout is delayed due to a much lower latent heat of evaporation, thus allowing for a much longer observation of the propagation path, which appears to be exponentially increasing in time. The experiments were preceded by our theoretical prediction of such a behavior, and this theory is also described in the present work. Due to the fact that this theory has been corroborated by the experimental data, the theory yields an adequate explanation and description of the CHF trigger and film boiling inception. In this second part of this report, model blister-like soft thermo-pneumatic artificial muscles with the embedded nanofibers impregnated with ethanol are developed. These muscles are based on pool boiling inside nanofiber mats.The muscles are essentially blister-like thermo-pneumatic soft actuators (BTSAs), which deflect in response to heat supplied to their bottom. The resulting deflections are on the scale of 1 cm, and the BTSAs are operational for several cycles. They are able to raise the artificial rigid scales, spines or fur/thin fibers attached to them emulating such animals as pangolin, hedgehog, and porcupine. They are also capable of removing the stickiest adhesive tapes attached to them, and thus hold great promise for biomedical applications where artificially grown skin patches should be removed from an underlying substrate without being damaged. The theory of the BTSA proposed in this work is in reasonable agreement with the acquired experimental data.
Bibliography Type:	Description: (Last Updated: 06/13/2025)
Articles in Peer-reviewed Journals	Staszel C, Yarin AL. "Exponential vaporization fronts and critical heat flux in pool boiling." Int J Heat Mass Transf. 2018 Nov;98:171-6. Epub 2018 Sep 26. https://doi.org/10.1016/j.icheatmasstransfer.2018.08.019 ; PubMed PMID: 30381775; PubMed Central PMCID: PMC6204265 , Nov-2018
Articles in Peer-reviewed Journals	An S, Kang DJ, Yarin AL. "A blister-like soft nano-textured thermo-pneumatic actuator as an artificial muscle." Nanoscale. 2018 Sep 13;10(35):16591-600. https://doi.org/10.1039/c8nr04181d ; PubMed PMID: 30152831; PubMed Central PMCID: PMC6250565 , Sep-2018